英文版 method

1. Experimental flow

1.1.Sample Quality Control

Please refer to QC report for methods of sample quality control.

1.2 DNA fragmentation (DNA Shearing)

The genomic DNA was randomly interrupted by a Covaris fragmentation device into fragments of about 350 bp.

1.3 End repair reaction (End Repair)

The fragmented DNA has a protrusion at the 5' or 3' end, and an end-complement system is added to the purified DNA fragments, in which the exonuclease activity of T4 DNA polymerase is used to digest the single-stranded protrusion at the 3' end, and its polymerase activity is used to complement the protrusion at the 5' end. At the same time phosphate kinase (PNK) adds a phosphate group to the 5' end, which is required for subsequent ligation reactions. After purification with Agencourt AMPure XP magnetic beads, a library of short, flat-ended DNA fragments containing a phosphate group at the 5' end is obtained.

1.4 Addition of poly(A) tail at 3' end (Adenlylate 3’ Ends)

The above system was supplemented with 3' end-addition poly(A) buffer reaction system. A single adenylate "A" is added to the 3' end of the end-modified double-stranded DNA, which is to prevent the flat ends of DNA fragments from self-associating with each other, so that they can be complementarily paired with the single "T" protrusion at the 5' end of the sequencing junction in the next step, and accurately connected, effectively reducing the crosstalk between library fragments themselves.

1.5 Adapter Ligation

Ligation buffer and double-stranded sequencing adapters were added to the above reaction system, T4 DNA Ligase was used to ligate Illumina sequencing adapters to both ends of the library DNA.

1.6 Size Selection

For libraries that have been attached to adapters, fragment size screening should be performed by the Agencourt SPRIselect kit (Beckman Coulter, USA, Catalog # : 2358413) at the same time as the library is purified. Double Size Selection was adopted,specifically, the SPRI beads were used to remove the small fragments on the left side of the target domain, and then the large fragments located on the right side of the target fragment region were removed, so that the original library with moderate fragment size was selected for the next step of PCR amplification.Excess sequencing adapters and adapter self-linkage products in the system have been removed from the purified library to avoid invalid amplification during the PCR process and to eliminate the impact on on-board sequencing.

1.7 PCR Amplification

The original library is amplified using a high-fidelity polymerase to ensure a sufficient total amount of library. In addition, this step is able to efficiently enrich for DNA fragments with adapters at both ends, which are the only fragments that can be amplified efficiently.With the premise of ensuring sufficient products, the bias introduced due to the excessive number of amplification cycles was reduced.The final concentration of each library was accurately determined using Qubit 3.0.

1.8 Library Quality Assessment

After the library construction was completed, the insert size of the library was detected by Agilent 5400 system (AATI), and after the insert size met the expectation, the effective concentration of the library (1.5 nM) was accurately quantified by qPCR to ensure the quality of the library.

1.9 Bridge PCR

When the library is qualified, the library is then sequenced on the Illumina Novaseq platform according to the validated concentration and data output of the library. Is the process of inoculthe captured library onto the FlowCell chip for amplification.The inner surface of the FlowCell channel has two different DNA primers that complement the adaptor sequences at both ends of the DNA library and are attached to the FlowCell as a covalent bond.The specific steps are described as follows:

a.The DNA library is added to the chip. Since the DNA sequence of the double end and the primer sequence on the chip are complementary, complementary hybridization will be performed. After hybridization, dNTP and polymerase are added. the polymerase starts from the primer, along the template, to synthesize a DNA strand complementary to the original DNA sequence;

b.NaOH solution is added to make the DNA duplex unchain, wash away the DNA chain that is not covalently connected to the chip, and retain the newly synthesized DNA chain that is covalently connected to the chip;

c. The neutralizing liquid is added to the liquid flow magnetic to neutralize the alkaline solution, then the other end of the DNA and the other primer on the chip undergo complementary hybridization, and the enzyme and dNTP are added to synthesize a new DNA chain. The alkali solution is added again to separate the two DNA strands, then the neutralisation solution is added, followed by the addition of the enzyme and dNTP and the synthesis of the new strand with the new primer. This process is repeated continuously, and the DNA strands grow exponentially.

1.10 Illumina platform for PE150 on-board sequencing

PE150 is Pair-end 150bp, high-throughput sequencing. In the constructed DNA small fragment library, each insert fragment was sequenced at both ends, each end was sequenced at 150bp, as follows.

After completion of bridge PCR, the synthesised double strand is turned into a single strand that can be sequenced;

a. A specific group of one of the primers on the chip is cut and the chip is rinsed with alkali solution, causing the DNA double strand to unravel and the cut DNA strand to be washed away, leaving the strand covalently bonded.
b. Add neutral solution, sequencing primers and fluorescently labelled dNTP, in which the four dNTPs are labelled with four different fluorescent labels and their 3' ends are blocked by azide groups. The dNTP is then synthesised into the new DNA strand by adding polymerase, which can only extend one base per cycle due to the blockage of the 3' end by the sodium azide group. After completing a cycle, the excess dNTP, enzyme, etc. are washed out and placed under a microscope for laser scanning to determine the type of newly synthesised base based on the fluorescence emitted, and the template base can be inferred through the principle of complementarity.
c. After completing the cycle, chemicals are added that allow the sodium azide group and the fluorescent group to be cleaved off, exposing the hydroxyl group at the 3' end. New dNTP and enzyme are added, another base is extended, and the excess dNTP and enzyme are flushed out before the round of microscopic laser scanning is performed to read out this round of bases. This cycle is repeated over and over again for hundreds of base reads.

2. Bioinformatic analysis

The raw sequences were analysed for information at the end of sequencing. The data quality was evaluated to determine whether it met the standard. If the data meets the standard, the samples will be tested for variants, including SNP, InDel, CNV, SV, and annotated with the corresponding variant information. If the data do not meet the quality standard, additional testing or library reconstruction is required according to the actual situation.

2.1 Data quality control

2.1.1 Raw data

The original fluorescence image files obtained from Illumina platform aretransformed to short reads (Raw data) by base calling and these short readsare recorded in FASTQ (Cock et al., 2010) format, which contains sequenceinformation and corresponding sequencing quality information.

2.1.2 Evaluation of data (Data quality control)

Sequence artifacts, including reads containing adapter contamination, low quality nucleotides and unrecognizable nucleotide (N), undoubtedly set the barrier for the subsequent reliable bioinformatics analysis. Hence quality controlis an essential step and applied to guarantee the meaningful downstream analysis.
The steps of data processing were as follows:
(1) Discard a paired reads if either one read contains adapter contamination(>10 nucleotides aligned to the adapter, allowing ≤ 10% mismatches);
(2) Discard a paired reads if more than 10% of bases are uncertain in eitherone read;
(3) Discard a paired reads if the proportion of low quality (Phred quality <5)bases is over 50% in either one read.3 References

2.1.3 Reads Mapping to Reference Sequence

Valid sequencing data were compared to the reference genome by BWA (Li et al., 2009a) software (parameter: mem -t 4 -k 32 -M), and the comparison results were removed from duplicates by SAMTOOLS (Li et al., 2009b) (parameter: rmdup).

2.2 Variant detection and annotation

2.2.1 SNP

SNPs (Single Nucleotide Polymorphisms) are mainly DNA sequence polymorphisms caused by variations in individual nucleotides at the genomic level, including transitions, inversions, etc. of individual bases.

The raw SNP sets are called by GATK(DePristo et al., 2011) with the parameters as ‘ – gcpHMM 10 -stand_emit_conf 10 -stand_call_conf 30’. Thenwe filtered this sets using the following criteria:

SNP: QD < 2.0, FS > 60.0, MQ < 30.0, HaplotypeScore > 13.0,
MappingQualityRankSum < -12.5, ReadPosRankSum < -8.0

2.2.2 InDel

InDel, known as Insertion and Deletion, refers to the insertion and deletion of small fragments. Insertions and deletions that occur either in the coding region or at the splice site may alter the translation of the protein. Shift mutations, in which the length of the inserted or missing base string is a non-integer multiple of 3, may result in a change in the entire reading frame; shift mutations have a greater impact on gene function than non-shift mutations and are subject to greater screening pressure in the population.

On the basis of the comparison results, GATK (-gcpHMM 10 -stand_emit_conf 10 -stand_call_conf30) detected small fragment insertions and deletions (InDel) less than 50bp in length. The following criteria were selected for filtering：

INDEL: QD < 2.0, FS > 200.0, ReadPosRankSum < -20.0

2.2.3 CNV detection

CNV (copy number variation) refers to an increase or decrease in the copy number of a genomic segment. It can be detected by CNVnator (parameter: -call 100) (Abyzov et al., 2011), i.e., determining potential deletion and duplication by the depth of coverage of different reads on the genome.

2.2.4 SV detection

SV (structural variation) refers to large insertions, deletions, inversions, translocations and other sequences at the genomic level. Insertions, deletions, inversions and translocations between samples and reference genomes can be detected by using BreakDancer (Chen et al., 2009) software, based on the pair-end reads comparison to the relationship above the reference genome and the actual Insert size. The following are some of the main features of PE reads: insertion (INS), deletion (DEL), inversion (INV), intra-chromosomal translocation (ITX), and inter-chromosomal translocation (CTX). Filtering to remove SV results with less than 2 support for PE reads

2.3 Annotation

ANNOVAR (Wang et al., 2010) was used for functional annotation of variants.The UCSC known genes were used for gene and region annotations.

3 References

Abyzov A, Urban AE, Snyder M, Gerstein M. CNVnator: an approach to discover, genotype,and characterize typical and atypical CNVs from family and population genomesequencing. Genome Res. 2011;21(6):974-984. doi:10.1101/gr.114876.110

Chen K, Wallis JW, McLellan MD, et al. BreakDancer: an algorithm for high-resolutionmapping of genomic structural variation. Nat Methods. 2009;6(9):677-681.doi:10.1038/nmeth.1363

Cock PJ, Fields CJ, Goto N, Heuer ML, Rice PM. The Sanger FASTQ file formatfor sequences with quality scores, and the Solexa/Illumina FASTQ variants.Nucleic Acids Res. 2010;38(6):1767-1771. doi:10.1093/nar/gkp1137

Li H, Durbin R. Fast and accurate short read alignment with Burrows-Wheelertransform.Bioinformatics.2009a;25(14):1754-1760.doi:10.1093/bioinformatics/btp324

Li H, Handsaker B, Wysoker A, et al. The Sequence Alignment/Map format andSAMtools.Bioinformatics.2009b;25(16):2078-2079.doi:10.1093/bioinformatics/btp3523

Wang K, Li M, Hakonarson H. ANNOVAR: functional annotation of geneticvariants from high-throughput sequencing data. Nucleic Acids Res.2010;38(16):e164. doi:10.1093/nar/gkq603